

Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2015–2025)(Exposé)

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1 Introduction

1.1 Problem Statement and Motivation

Epilepsy, a chronic neurological disorder affecting 60 million people worldwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveinson et al., 2020 found that 69% of Deaths due to a Generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.

While reliable prediction is already possible through electroencephalography (EEG), its operational complexity render it unsuitable for ambulatory use. To address the need for ambulatory detection many are experimenting with utilizing autonomic and motion signals to detect and predict seizures.

However, translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the intricate and non-linear complex temporal dynamic of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction uniquely addressing these limitations.

This review therefore synthesizes and critically evaluates recent advances in time-series deep learning—particularly convolutional neural networks (CNNs) and long short term memory (LSTMs)—applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of how architectural choices, training paradigms (e.g., transfer learning, personalization), and rigorous, prospective evaluation can converge to meet the stringent clinical requirements of high sensitivity and minimal false alarm rates (FAR) in real-world deployment.

2 Theoretical Foundations for Ambulatory Seizure Monitoring

2.1 The Ambulatory Monitoring Paradigm: Requirements and Challenges

The translation of seizure detection from the controlled clinical environment to an ambulatory setting redefines approaches and performance requirements. In clinical environments maximum sensitivity is prioritized over FAR, when engineering for ambulatory use a balance of both metrics is needed. Failing to detect a seizure (low sensitivity) poses a safety risk, While excessive false alarms (high FAR) leads to desensitizing and "alarm fatigue" and the device being ignored or not used at all. While more complex models can find more intricate patterns they also introduce the "black-box" problem wherein the model lacks Transparency. Therefore the design should incorporate explanations for decisions to ensure users trust and regulatory compliance.

2.2 Physiological Biosignals for Wearable Sensing

Wearable devices circumvent the impracticality of EEG by capturing peripheral biosignals that serve as proxies for central nervous system activity during seizures. These proxies will inherently relay incomplete and noisy signals so a combination of multiple sensors is required. The acceleration/attitude sensor excels at detecting muscle convulsions but is noisy in daily use, as it captures all movement. The electrodermal activity (EDA) sensor detects autonomic arousal by measuring the skin conductance associated with sweat gland activity.

2.3 Seizure Manifestations and Corresponding Biosignals

In ambulatory monitoring, wearable devices rely on a fundamental proxy relationship: peripheral biosignals reflect central nervous system activity. Different seizure types therefore exhibit unique physiological patterns across these proxy signals.

Generalized tonic-clonic seizures, due to their high risk profile, are a primary target for ambulatory detection. They produce a stereotypic pattern across multiple biosignals: violent, rhythmic motion (detectable via accelerometry/ACC) coinciding with a pronounced autonomic storm marked by sustained tachycardia (visible in electrocardiography/ECG or photoplethysmography/PPG) and a sharp rise in sympathetic tone (measured as increased skin conductance via electrodermal activity/EDA). In contrast, focal non-motor seizures may manifest only through subtler autonomic shifts, such as a change in heart rate variability (HRV, derived from ECG/PPG) or isolated EDA peaks, with ACC serving primarily to contextualize other signals

2.4 Seizure Manifestations and Corresponding Biosignals

Wearable biosignals serve as essential proxies for cerebral activity during seizures, with different seizure types producing distinct patterns across these peripheral signals. This proxy relationship forms the detection logic for ambulatory monitoring: specific seizure semiology dictates which combination of physiological signals presents the most reliable signature for algorithmic identification. The motor component of seizures, particularly the convulsive movements of generalized tonic-clonic (GTC) events, is captured by accelerometry (ACC). ACC data records characteristic movement patterns and posture changes, with specific preprocessing techniques extracting attitude-angle signatures that distinguish convulsive activity from daily motion (Wang et al., 2025; Wu et al., 2024). Autonomic manifestations, which occur across most seizure types, are reflected in multiple complementary biosignals. Electrodermal activity (EDA) directly measures sympathetic nervous system output through skin conductance, with phasic skin conductance responses (SCRs) indicating sympathetic arousal during peri-ictal states (Jain et al., 2017). Cardiac activity provides perhaps the most studied autonomic proxy: the electrocardiogram (ECG) captures direct cardiac data where ictal tachycardia and other morphological changes are well-documented seizure markers (Leal et al., 2017). Heart rate variability (HRV), derived from ECG or photoplethysmography (PPG), quantifies autonomic balance through time-domain, frequency-domain, and non-linear measures, reflecting central autonomic network modulation during seizures (Mason et al., 2024; Pavei et al., 2017). PPG offers a practical wearable alternative for capturing pulse-rate and pulse-rate variability, though with greater susceptibility to motion artifact than ECG (Seth et al., 2023).

This mapping between seizure phenotype and its corresponding proxy signatures establishes the physiological basis upon which detection algorithms operate, guiding both sensor selection and multimodal fusion strategies for ambulatory monitoring systems.

2.5 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles). The methodological approach uses a multi-stage search and transparent extraction/synthesis procedures as outlined below.

1. Search strategy (Multi-stage, 2015–2025):

- *Scoping*: Google Scholar for broad coverage and identifying candidate papers.
- *Formal queries*: IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.

- *Citation chasing*: Backward and forward citation tracking on key papers to ensure completeness.

2. Inclusion / Exclusion criteria:

- *Inclusion*: Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
- *Exclusion*: EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate a concept matrix with the following fields:

1. Setting: clinical vs ambulatory.
2. Modality: ECG/HRV, PPG, EDA, ACC.
3. Task: detection vs prediction.
4. Model class and fusion strategy.
5. Study phase: retrospective, pseudo-prospective, prospective.
6. Evaluation granularity: sample- vs alarm-based.
7. Windowing specifics (window length, stride, overlap).
8. Metrics: sensitivity, FAR/h, AUC, f1-score, latency.
9. Personalization.
10. Interpretability and safety considerations.
11. Data characteristics and dataset identity.
12. Missing data handling (imputation, exclusion) and synthetic data use.
13. Inference requirements (latency, computational load, hardware).

Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

2.6 Provisional Table of Contents

Target Length: Approx. 15 Pages of Text

1. **Introduction** (approx. 1.5 pages)
 - 1.1 Problem Statement and Motivation
 - 1.2 Research Aim and Questions
 - 1.3 Structure of this Review
2. **Methodological Approach** (approx. 1.5 pages)

- 2.1 Search Strategy (Multi-stage, 2015–2025)
- 2.2 Inclusion and Exclusion Criteria
- 2.3 Extraction Dimensions and Concept Matrix
- 2.4 PRISMA Flow Diagram

3. Theoretical Background and Clinical Concepts (approx. 2 pages)

- 3.1 Clinical Deployment Requirements (Sensitivity vs. FAR/h, Alarm Fatigue)
- 3.2 Seizure Classification and Clinical Implications
- 3.3 Autonomic Markers and Wearable Sensing (ECG/HRV, PPG, EDA, ACC)
- 3.4 Study Phases: Retrospective vs. Pseudo-Prospective vs. Prospective

4. Modeling Landscape and Design Dimensions (approx. 5 pages)

- 4.1 Architectural Evolution: Feature Engineering vs. Deep Learning
 - Traditional Pipelines (SVM, RF, kNN with HRV/EDA Features)
 - Deep Temporal Models (1D-CNN, LSTM, TCN)
 - Attention and Transformer Variants
- 4.2 Sensor Modalities and Fusion Strategies
 - Feature-Level, Decision-Level, and Representation-Level Fusion
 - Complementary Cues under Noise and Non-Stationarity
- 4.3 Personalization and Domain Shift
 - Patient-Specific Fine-Tuning and Calibration
 - Cross-Dataset Evaluation and External Validation
- 4.4 Robustness and Generalization
 - Artifact Handling and Augmentation Strategies
 - Out-of-Distribution Detection
- 4.5 Deployment Constraints (Latency, Energy, Interpretability)

5. Results: Synthesis and Comparative Analysis (approx. 3 pages)

- 5.1 Performance Ranges by Study Phase and Setting
- 5.2 Model Comparison: Feature-Based vs. Deep Learning
- 5.3 Impact of Temporal Choices (Window Length, Stride, Context)
- 5.4 Evidence on Robustness and Domain Shift
- 5.5 Dataset Gaps, Class Imbalance, and Augmentation Effects

6. Discussion (approx. 1.5 pages)

- 6.1 Trade-offs: Complexity vs. Latency vs. Interpretability
- 6.2 Clinical vs. Ambulatory Deployment Considerations
- 6.3 Limitations of Current Evidence

7. Conclusion and Future Work (approx. 0.5 pages)

- 7.1 Summary of Key Findings
- 7.2 Practical Recommendations
- 7.3 Research Gaps and Future Directions

Note: The following sections provide preliminary background to contextualize the research questions. They will be expanded with systematic literature findings in the final paper.

3 Modeling Landscape and Design Dimensions

Detection and preictal prediction from ECG/HRV, PPG, EDA, and ACC are framed by windowing, labeling, and alarm logic, with alarm-based sensitivity, FAR/h, and latency as the outcomes used in the research questions.

3.1 Architectural Evolution: Feature Engineering vs. Deep Learning

Classical pipelines utilizing HRV, EDA, and ACC features combined with SVM, RF, or kNN remain competitive in retrospective settings (Mason et al., 2024; Seth et al., 2023). However, deep temporal models such as 1D-CNN, LSTM, and TCN often improve phase discrimination. Recently, attention variants and Transformers are being explored on these signals (Abtahi et al., 2025; Kalousios et al., 2024; Pordoy et al., 2025; Yang et al., 2022). A key inquiry of this review is determining how feature-based models compare to deep models and whether attention mechanisms add distinct value under ambulatory, alarm-based evaluation.

3.2 Sensor Modalities and Fusion Strategies

Fusion across ECG, EDA, ACC, and PPG at the feature, decision, or representation level seeks complementary cues and is relevant to maintaining performance under noise and non-stationarity (Pordoy et al., 2025; Yang et al., 2022).

3.3 Personalization and Domain Shift

Patient-specific fine-tuning and post-hoc calibration shift the operating point toward lower FAR/h and higher sensitivity in ambulatory use (Andrade et al., 2024). Thresholding and periodic recalibration are commonly used to handle drift.

3.4 Robustness and Generalization

Robustness requires explicit checks of domain shift across patient, device, site, and activity. Cross-dataset evaluation (e.g., training on TUH and testing on RPAH or EPILEP-SIAE) exposes this shift and favors alarm-based over sample-based metrics (Andrade et al., 2024; Yang et al., 2022). Artifact handling, augmentation through time-warping, jitter, or scaling, and Out-of-Distribution (OOD) detection support generalization (Kalousios et al., 2024; Seth et al., 2023).

3.5 Deployment Constraints: Latency, Energy, and Interpretability

Wearable seizure detection imposes strict deployment constraints. Detection latency must remain below clinically relevant thresholds (typically seconds to allow caregiver response or patient warning), which limits model complexity and inference time. Energy consumption determines battery life; continuous inference on resource-constrained MCUs or mobile processors requires lightweight architectures such as quantized CNNs or efficient RNNs. Compact sequence models (e.g., depthwise-separable 1D-CNNs, temporal convolutional networks) and event-driven or on-demand processing reduce computational burden.

Interpretability remains critical for clinical adoption. Regulators and clinicians require transparency in alarm generation to trust automated decisions affecting patient safety. Post-hoc explanation methods (e.g., attention weights, Grad-CAM on temporal segments, SHAP values for HRV features) can highlight which signal segments or features triggered an alarm. Hybrid pipelines that combine interpretable HRV/EDA features with deep representations offer a middle ground between performance and explainability.

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